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Open Geospatial Consortium

OGC API - TILES (OPENAPI-DRIVEN SECTIONS SAMPLE)

STANDARD
Implementation

DRAFT

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I SECURITY CONSIDERATIONS

No security considerations have been made for this document.

1 SCOPE

1 SCOPE

This sample demonstrates the paths-bearing OpenAPI corpus: the normative sections below are generated from the bundled OGC API — Tiles OpenAPI 3.0 definition (copied verbatim from the opengeospatial/ogcapi-tiles release artifacts) through metanorma-plugin-openapi and lutaml-openapi, exercising the 3.0 normalization path and the Paths renderer.

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OGC API — TILES

2 OGC API — TILES

API version: 3.0.0 Example API Definition for OGC API — Tiles — Part 1: Core

3 SCHEMAS

3 SCHEMAS

3.1. confClasses

Type: object

Table 1 — Properties of confClasses

NAME	TYPE	DESCRIPTION
conformsTo	array	

3.2. link

Type: object

Table 2 — Properties of link

NAME	TYPE	DESCRIPTION
href	string	Supplies the URI to a remote resource (or resource fragment).
hreflang	string	A hint indicating what the language of the result of dereferencing the link should be.
length	integer	
rel	string	The type or semantics of the relation.
templated	boolean	This flag set to true if the link is a URL template.
title	string	Used to label the destination of a link such that it can be used as a human-readable identifier.
type	string	A hint indicating what the media type of the result of dereferencing the link should be.

NAME	TYPE	DESCRIPTION
varBase	string	A base path to retrieve semantic information about the variables used in URL template.

3.3. landingPage

Type: object

Table 3 — Properties of landingPage

NAME	TYPE	DESCRIPTION
attribution	string	The attribution should be short and intended for presentation to a user, for example, in a corner of a map. Parts of the text can be links to other resources if additional information is needed. The string can include HTML markup.
description	string	
links	array	
title	string	While a title is not required, implementors are strongly advised to include one.

3.4. exception

JSON schema for exceptions based on RFC 7807

Type: object

Table 4 — Properties of exception

NAME	TYPE	DESCRIPTION
detail	string	
instance	string	

NAME	TYPE	DESCRIPTION
status	integer	
title	string	
type	string	

3.5. collections

Type: object

Table 5 — Properties of collections

NAME	TYPE	DESCRIPTION
collections	array	
links	array	
numberMatched	integer	
numberReturned	integer	
timeStamp	string	

3.6. collectionInfo

Type: object

Table 6 — Properties of collectionInfo

NAME	TYPE	DESCRIPTION
crs	array	the list of coordinate reference systems supported by the API; the first item is the default coordinate reference system

NAME	TYPE	DESCRIPTION
dataType	``	
description	string	a description of the data in the collection
extent	``	
geometryDimension	integer	The geometry dimension of the features shown in this layer (0: points, 1: curves, 2: surfaces, 3: solids), unspecified: mixed or unknown
id	string	identifier of the collection used, for example, in URIs
itemType	string	indicator about the type of the items in the collection if the collection has an accessible /collections/{collectionId}/items endpoint
links	array	
maxCellSize	number	Maximum cell size for usage of the collection
maxScale Denominator	number	Maximum scale denominator for usage of the collection
minCellSize	number	Minimum cell size for usage of the collection
minScale Denominator	number	Minimum scale denominator for usage of the collection
title	string	human readable title of the collection

3.7. extent

The extent of the data in the collection. In the Core only spatial and temporal extents are specified. Extensions may add additional members to represent other extents, for example, thermal or pressure ranges.

The first item in the array describes the overall extent of the data. All subsequent items describe more precise extents, e.g., to identify clusters of data. Clients only interested in the overall extent will only need to access the first item in each array.

Type: object

Table 7 — Properties of extent

NAME	TYPE	DESCRIPTION
spatial	object	The spatial extent of the data in the collection.
temporal	object	The temporal extent of the features in the collection.

3.8. extent-uad

The extent module only addresses spatial and temporal extents. This module extends extent by specifying how intervals and crs properties can be used to specify additional geometries.

3.9. crs

3.10. dataType

3.11. timeStamp

This property indicates the time and date when the response was generated using RFC 3339 notation.

Type: string

3.12. numberReturned

The number of features in the feature collection. A server may omit this information in a response, if the information about the number of features is not known or difficult to compute. If the value is provided, the value shall be identical to the number of items in the “features” array.

Type: integer

3.13. numberMatched

The number of features of the feature type that match the selection parameters like bbox.

Type: integer

3.14. tileSet

A resource describing a tileset based on the OGC TileSet Metadata Standard. At least one of the ‘TileMatrixSet’, or a link with ‘rel’ http://www.opengis.net/def/rel/ogc/1.0/tiling-scheme

Type: object

Table 8 — Properties of tileSet

NAME	TYPE	DESCRIPTION
accessConstraints	string	Restrictions on the availability of the Tile Set that the user needs to be aware of before using or redistributing the Tile Set
attribution	string	Short reference to recognize the author or provider
boundingBox	``	
centerPoint	``	
created	``	
crs	``	

NAME	TYPE	DESCRIPTION
dataType	``	
description	string	Brief narrative description of this tile set
epoch	number	Epoch of the Coordinate Reference System (CRS)
keywords	array	keywords about this tileset
layers	array	
license	string	License applicable to the tiles
links	array	Links to related resources. Possible link 'rel' values are: 'http://www.opengis.net/def/rel/ogc/1.0/dataset' for a URL pointing to the dataset, 'item' for a URL template to get a tile; 'alternate' for a URL pointing to another representation of the TileSetMetadata (e.g a TileJSON file); 'http://www.opengis.net/def/rel/ogc/1.0/tiling-scheme' for a definition of the TileMatrixSet; 'http://www.opengis.net/def/rel/ogc/1.0/geodata' for pointing to a single collection (if the tileset represents a single collection)
mediaTypes	array	Media types available for the tiles
pointOfContact	string	Useful information to contact the authors or custodians for the Tile Set
style	``	
tileMatrixSetLimits	array	Limits for the TileRow and TileCol values for each TileMatrix in the tileMatrixSet. If missing, there are no limits other that the ones imposed by the TileMatrixSet. If present the TileMatrices listed are limited and the rest not available at all
tileMatrixSetURI	string	Reference to a Tile Matrix Set on an offical source for Tile Matrix Sets such as the OGC NA definition server (http://www.opengis.net/def/tms/). Required if the tile matrix set is registered on an open official source.
title	string	A title for this tileset
updated	``	
version	string	Version of the Tile Set. Changes if the data behind the tiles has been changed

3.15. tileSet-item

A minimal tileset element for use within a list of tilesets linking to full description of those tilesets.

Type: object

Table 9 — Properties of tileSet-item

NAME	TYPE	DESCRIPTION
crs	``	
dataType	``	
links	array	Links to related resources. A 'self' link to the tileset as well as a 'http://www.opengis.net/def/rel/ogc/1.0/tiling-scheme' link to a definition of the TileMatrix Set are required.
tileMatrixSetURI	string	Reference to a Tile Matrix Set on an offical source for Tile Matrix Sets such as the OGC NA definition server (http://www.opengis.net/def/tms/). Required if the tile matrix set is registered on an open official source.
title	string	A title for this tileset

3.16. tileMatrixSet

A definition of a tile matrix set following the Tile Matrix Set standard. For tileset metadata, such a description (in tileMatrixSet property) is only required for offline use, as an alternative to a link with a http://www.opengis.net/def/rel/ogc/1.0/tiling-scheme relation type.

Type: object

Table 10 — Properties of tileMatrixSet

NAME	TYPE	DESCRIPTION
boundingBox	``	
crs	``	

NAME	TYPE	DESCRIPTION
description	string	Brief narrative description of this tile matrix set, normally available for display to a human
id	string	Tile matrix set identifier. Implementation of 'identifier'
keywords	array	Unordered list of one or more commonly used or formalized word(s) or phrase(s) used to describe this tile matrix set
orderedAxes	array	
tileMatrices	array	Describes scale levels and its tile matrices
title	string	Title of this tile matrix set, normally used for display to a human
uri	string	Reference to an official source for this tileMatrixSet
wellKnownScaleSet	string	Reference to a well-known scale set

3.17. tileMatrixSet-item

A minimal tile matrix set element for use within a list of tile matrix sets linking to a full definition.

Type: object

Table 11 — Properties of tileMatrixSet-item

NAME	TYPE	DESCRIPTION
crs	``	
id	string	Optional local tile matrix set identifier, e.g. for use as unspecified {tileMatrixSetId} parameter. Implementation of 'identifier'
links	array	Links to related resources. A 'self' link to the tile matrix set definition is required.
title	string	Title of this tile matrix set, normally used for display to a human
uri	string	Reference to an official source for this tileMatrixSet

3.18. tileMatrixLimits

The limits for an individual tile matrix of a TileSet's TileMatrixSet, as defined in the OGC 2D TileMatrixSet and TileSet Metadata Standard

Type: object

Table 12 — Properties of tileMatrixLimits

NAME	TYPE	DESCRIPTION
maxTileCol	integer	
maxTileRow	integer	
minTileCol	integer	
minTileRow	integer	
tileMatrix	string	

3.19. 2DPoint

A 2D Point in the CRS indicated elsewhere

Type: array

3.20. 2DBoundingBox

Minimum bounding rectangle surrounding a 2D resource in the CRS indicated elsewhere

Type: object

Table 13 — Properties of 2DBoundingBox

NAME	TYPE	DESCRIPTION
crs	``	
lowerLeft	``	
orderedAxes	array	
upperRight	``	

3.21. FeatureCollection

Type: object

Table 14 — Properties of FeatureCollection

NAME	TYPE	DESCRIPTION
bbox	array	
features	array	
type	string	

3.22. enumeration

Type: object

Table 15 — Properties of enumeration

NAME	TYPE	DESCRIPTION
enum	array	

NAME	TYPE	DESCRIPTION
type	string	

3.23. all-collections

Type: string

3.24. vectorTiles-collections

Type: string

3.25. coverage-collections

Type: string

3.26. styles

Type: string

3.27. tileMatrixSets

Type: string

4 PATHS

4 PATHS

4.1. GET /

Retrieve the OGC API landing page for this service. === GET /api

Retrieve this API definition. === GET /api/all-collections

Retrieve the list of collections available from this API implementation & deployment. === GET /api/coverage-collections

Retrieve the list of coverages collections available from this API implementation & deployment. === GET /api/styles

Retrieve the list of dataset styles available from this API implementation & deployment. === GET /api/tileMatrixSets

Retrieve the list of shared TileMatrixSets available from this API implementation & deployment. === GET /api/vectorTiles-collections

Retrieve the list of collections supporting vector tiles available from this API implementation & deployment. === GET /collections

Retrieve the list of geospatial data collections available from this service. === GET /collections/{collectionId}

Retrieve the description of a collection available from this service. === GET /collections/{collectionId}/coverage/tiles

Retrieve the list of available coverage tilesets for the specified collection. === GET /collections/{collectionId}/coverage/tiles/{tileMatrixSetId}

Retrieve the coverage tileset metadata for the specified collection and tiling scheme (tile matrix set) === GET /collections/{collectionId}/coverage/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve coverage tiles === GET /collections/{collectionId}/map/tiles

Retrieve a list of all map tilesets for specified collection. === GET /collections/{collectionId}/map/tiles/{tileMatrixSetId}

Retrieve a map tile set metadata for the specified collection and tiling scheme (tile matrix set) === GET /collections/{collectionId}/map/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve a map tile from the specified collection === GET /collections/{collectionId}/styles/{styleId}/map/tiles

Retrieve a list of styled map tilesets for the specified collection === GET /collections/{collectionId}/styles/{styleId}/map/tiles/{tileMatrixSetId}

Retrieve the map tileset metadata for the specified collection, style and tiling scheme (tile matrix set). === GET /collections/{collectionId}/styles/{styleId}/map/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve a map tile for a specified collection and style === GET /collections/{collectionId}/styles/{styleId}/tiles

Retrieve a list of vector tilesets for the specified collection intended for use with a specified style === GET /collections/{collectionId}/styles/{styleId}/tiles/{tileMatrixSetId}

Retrieve the vector tileset metadata for the specified collection, style and tiling scheme (tile matrix set). === GET /collections/{collectionId}/styles/{styleId}/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve vector tiles for a specified collection, intended for use with a specified style === GET /collections/{collectionId}/tiles

Retrieve a list of available vector tilesets for the specified collection. === GET /collections/{collectionId}/tiles/{tileMatrixSetId}

Retrieve the vector tileset metadata for the specified collection and tiling scheme (tile matrix set) === GET /collections/{collectionId}/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve a vector tile from a collection. === GET /conformance

Retrieve the set of OGC API conformance classes that are supported by this service. === GET /map/tiles

Retrieve the list of all default map tilesets for the whole dataset === GET /map/tiles/{tileMatrixSetId}

Retrieve a default map tileset of the whole dataset for the specified tiling scheme (tile matrix set) === GET /map/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve a default map tile of the whole dataset === GET /styles/{styleId}/map/tiles

Retrieve the list of styled map tilesets for the whole dataset === GET /styles/{styleId}/map/tiles/{tileMatrixSetId}

Retrieve a styled map tileset of the whole dataset for the specified tiling scheme (tile matrix set) === GET /styles/{styleId}/map/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve a styled map tiles === GET /styles/{styleId}/tiles

Retrieve the list of vector tilesets intended for use with a specified style for the whole dataset === GET /styles/{styleId}/tiles/{tileMatrixSetId}

Retrieve a vector tileset intended for use with a specified style of the whole dataset for the specified tiling scheme (tile matrix set) === GET /styles/{styleId}/tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve vector tiles intended for use with a specified style === GET /tileMatrixSets

Retrieve the list of available tiling schemes (tile matrix sets) === GET /tileMatrixSets/{tileMatrixSetId}

Retrieve the definition of the specified tiling scheme (tile matrix set) === GET /tiles

Retrieve a list of available vector tilesets for the dataset === GET /tiles/{tileMatrixSetId}

Retrieve the vector tileset metadata for the whole dataset and the specified tiling scheme (tile matrix set) === GET /tiles/{tileMatrixSetId}/{tileMatrix}/{tileRow}/{tileCol}

Retrieve a vector tile including one or more collections from the dataset.